

UDAN-STAT

Unified Domestic Airfare & Network Sovereign Tariff Analytics Platform

SIH 2026 Idea Submission

Complete System & Dashboard Dossier

React 19 • FastAPI • SQLite/PostgreSQL

Sub-45ms Verified Query Latency

Executive Summary & System Purpose

UDAN-STAT is a sovereign regulatory intelligence platform engineered for the Ministry of Civil Aviation (MoCA) and DGCA to eliminate regulatory blindspots in Indian domestic aviation. It replaces arbitrary manual spot-checks with an empirical, tamper-proof mathematical framework:

- **National Airfare Price Index:** Modified Laspeyres index dynamically weighted by DGCA quarterly Form A/B passenger matrices across 80 scheduled domestic corridors covering ~92% scheduled capacity.
- **Statutory UDAN RCS Auditor:** First-of-its-kind Geodesic distance auditor benchmarking regional fares against MoCA statutory caps (₹2,250 – ₹5,800) with 1-click Show-Cause Notice dispatching under Rule 135 of Aircraft Rules, 1937.
- **Antitrust & Macroeconomic Simulation:** HHI monopoly detection and Aviation Turbine Fuel (ATF) shock elasticity modeling safeguarding ₹4,500+ Cr of consumer welfare.

1. Executive Landing & Intelligence Overview

Route: /

Primary Objective: Acts as the ministerial executive gate providing senior policymakers with an instantaneous 10-second pulse of national airfare inflation and surge velocity.

KEY VISUAL COMPONENTS

- **Sovereign Flash KPIs:** National Benchmark Index (124.50), 24h Surge Velocity, 80 Corridors, 92% Domestic Seat Capacity.
- **Multi-Horizon Carousel:** Compares T+1 emergency pricing against T+30/T+45 advance purchase tiers.
- **Sovereign Module Grid:** 7 direct navigational pathways into analytical sub-engines.

UNDERLYING MECHANICS

- Aggregates weighted median pricing across all active carrier feeds.
- Translates complex statistical spreads into high-level ministerial alerts.

Jury Defense Line: "Sir, the Landing Cockpit replaces subjective complaints with an objective, real-time national indicator of domestic airfare inflation."

2. National Airfare Price Index Calculator

Route: /dashboard

Primary Objective: The computational core computing India's official weighted Laspeyres domestic airfare price index.

KEY VISUAL COMPONENTS

- **Flight Control Panel:** Cabin class filter (Economy / Business) and 5 Active Fleet toggles (6E, AI, SG, QP, IX).
- **Lead-Horizon Spreads:** T+1 (+50.8%), T+7 (+24.5%), T+15 (+18.8%), T+30 (+11.5%), T+45 (+6.1%).
- **Plotly 30-Day Forward Trajectory:** Historical vs forward price index projection against Base 100.00 parity.
- **Route Network Comparison:** Metro trunks vs Tier-2 regional network indices.

MATHEMATICAL FORMULA

$$APIx_t = \sum [w_r \times (P(r,t) / P(r,0))] \times 100$$

Where w_r is DGCA quarterly passenger share, $P(r,t)$ is clean median fare, and Base Sept 2022 = 100.00.

Jury Defense Line: "We do not take raw averages. The index is passenger-weighted via DGCA Form A/B so high-volume trunk routes accurately drive the macroeconomic score."

3. Mathematical & Statistical Methodology

Route: /methodology

Primary Objective: Mathematical defense engine demonstrating data sanitization and proof against arithmetic mean skew.

KEY VISUAL COMPONENTS

- **Interactive IQR Outlier Sanitizer:** Live tool calculating Q1 (25th %), Q3 (75th %), IQR, and Upper Bound.
- **Mean vs Median Scatter Comparison:** Proof showing how 1 predatory ₹18,500 ticket skews arithmetic mean by 42% while Median remains truthful.
- **80-Route Passenger Weight Ledger:** Full DGCA normalized passenger share table ($\sum w_r = 1.0$).

TWO-STAGE STATISTICAL DEFENSE

- **Stage 1:** Trim unrepresentative fares outside $[Q1 - 1.5 \cdot IQR, Q3 + 1.5 \cdot IQR]$.
- **Stage 2:** Compute robust statistical median $P(r,t)$ across clean distribution.

Jury Defense Line: "Sir, statistical IQR bounds eliminate promotional flash sales and predatory surge spikes before computing medians, guaranteeing mathematical integrity."

4. Antitrust & HHI Monopoly Cockpit

Route: /analysts

Primary Objective: Regulatory surveillance tool for the Competition Commission of India (CCI) to detect route monopolies and cartelization.

KEY VISUAL COMPONENTS

- **Herfindahl-Hirschman (HHI) Scorecard:** Color-coded market concentration status across 80 corridors.
- **Concentration Thresholds:** <1500 (Competitive), 1500-2500 (Moderate), >2500 (High Monopoly Risk).
- **Duopoly Radar:** Identifies corridors dominated by IndiGo + Air India (~90% combined market share).

REGULATORY FOCUS

- Flags artificial capacity throttling (ASKM drops) preceding holiday fare surges.
- Equips regulators to prevent predatory monopoly pricing.

Jury Defense Line: "HHI monitoring allows DGCA and CCI to proactively intervene on monopoly routes before passengers are subjected to exploitative pricing."

5. UDAN RCS Statutory Fare Cap Compliance Auditor

Route: /udan-rcs

Primary Objective: Flagship statutory engine monitoring regional corridors against distance-tiered legal fare caps mandated under the UDAN (Ude Desh Ka Aam Nagrik) Scheme.

KEY FEATURES & METRICS

- **4 Regulatory KPIs:** Monitored Corridors (62), Statutory Compliance Rate (%), Active Breaches Count, Total Excess Surcharge (₹1,76,738+).
- **Plotly Step-Line Chart:** MoCA distance ceiling step-line vs live observed carrier airfares.
- **VGF Clawback Estimator:** Calculates public subsidy recovery from airlines violating fare caps.

1-CLICK LEGAL NOTICE & EXPORT

- **DGCA Show-Cause Notice:** 1-Click modal auto-drafting official legal notice citing *Rule 135 of Aircraft Rules, 1937* with exact route evidence.
- **CSV Dossier Export:** Instant download of MoCA regulatory audit report.

Stage Length (Distance Tier)	Statutory RCS Fare Cap	Compliance Rule	Enforcement Action
≤ 250 km	₹2,250	Actual Fare ≤ Cap	Tag COMPLIANT (Green)
251 – 350 km	₹2,750	Actual Fare ≤ Cap × 1.10	Tag MARGINAL (Warning)
351 – 500 km	₹3,300	Actual Fare > Cap × 1.10	Tag BREACH → Calculate VGF Clawback
501 – 650 km	₹3,800	Haversine Great-Circle	Generate Rule 135 Show-Cause Notice
651 – 850 km	₹4,400	Airport GPS Lat/Long	Flag to DGCA Tariff Cell
851 – 1,100 km	₹5,100	Single Seat Outlier Filtered	Clawback Viability Gap Funding
> 1,100 km	₹5,800	Economy Class Strict Filter	Export Regulatory Audit Dossier

Jury Defense Line: "Sir, this feature makes our solution truly unique. It protects taxpayers' ₹1,000+ Crore Viability Gap Funding by auditing whether airlines are charging predatory rates on subsidized UDAN routes."

6. ATF Fuel Shock Elasticity Simulator

Route: /simulation

Primary Objective: Macroeconomic cost pass-through simulator modeling the financial impact of crude oil and jet fuel volatility on ticket prices.

KEY VISUAL COMPONENTS

- **Fuel Fluctuation Slider:** Interactive slider from -30% to +50% crude price shocks.
- **Elasticity Toggles:** Low ($\eta = 0.40$), Medium ($\eta = 0.65$), High ($\eta = 0.85$).
- **Consumer Financial Exposure Metric:** Quantifies national consumer burden in ₹ Crores.
- **Route Sensitivity Matrix:** Plots fuel sensitivity across 10 representative trunk corridors.

ECONOMIC RATIONALE

- ATF constitutes 35–40% of airline operational costs in India.
- Models legitimate fuel surcharge adjustments vs illegitimate predatory gouging.

Jury Defense Line: "When global crude jumps 20%, our simulator tells regulators exactly how much fare increase is mathematically justified versus artificial price hiking."

7. Carrier Fleet & Capacity Intelligence

Route: /fleet

Primary Objective: Operational aircraft inventory, fuel efficiency benchmarks, and capacity allocation across scheduled domestic carriers.

KEY VISUAL COMPONENTS

- **Carrier Profiles:** IndiGo (A320neo, A321neo), Air India (A350, B777), SpiceJet (B737, Q400), Akasa Air (B737 MAX 8), AI Express.
- **Seat Density vs Fares:** Single-class high-density vs full-service dual-class yield comparison.
- **Fuel Burn Benchmarks:** LEAP-1A / GTF engine efficiency differentials.

OPERATIONAL CONTEXT

- Correlates aircraft seating capacity with route-level yield curves.

Jury Defense Line: "Fleet intelligence tracks how aircraft upgrades directly alter an airline's baseline cost structure and pricing dynamics."

8. Regulatory Citations & Standards

Route: </references>

Primary Objective: Documents legal anchors, statistical guidelines, and Survey of India cartographic compliance.

- **Statutory Mandate:** Rule 135, Aircraft Rules (1937) — Tariff filing & prevention of predatory pricing.
- **Official Data Feeds:** DGCA Monthly Traffic Form A/B, MOSPI CPI Transport Sub-Index Manual, ICAO Doc 9626.
- **Sovereign Cartography:** 100% Survey of India (Sol) compliant territorial boundaries.

9. Sovereign Gov API Gateway

Route: </gov-api>

Primary Objective: Enterprise integration portal for National Informatics Centre (NIC) and MoCA automated systems.

- **REST Endpoints:** </api/index>, </api/udan/rcs-compliance>, </api/udan/export-rcs-report>.
- **Performance SLA:** Verified sub-45ms response latency with OpenAPI 3.0 Swagger specifications.

3-Minute Master Pitch for Judges & Evaluators

1. **The Problem:** Under the guise of dynamic pricing, airlines exploit emergency travelers and regional routes with 2.14x surge spikes while the government lacked a reproducible real-time price index.
2. **The Innovation (UDAN-STAT):** We created India's first Laspeyres airfare index weighted by DGCA Form A/B passenger distributions across 80 corridors, backed by statistical IQR outlier filtering.
3. **The Differentiator (UDAN RCS Auditor):** Our Haversine geodesic engine monitors regional routes against statutory fare caps (₹2,250–₹5,800), detects Viability Gap Funding (VGF) subsidy clawbacks, and auto-generates legal notices under Rule 135 of Aircraft Rules, 1937.
4. **Readiness:** 100% sovereign open-source stack (React 19 + FastAPI + SQLite/Postgres), live deployed on Vercel with zero commercial API licensing overhead.